

# Final Project – Individual Contribution Statement

Agri-DINO: Domain-Adaptive Self-Supervision for Fine-Grained Plant Recognition

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I contributed equally to the project. My primary responsibility was the representation-learning pipeline and the first-pass manuscript draft.

## Code and experiments

- Audited and cleaned the 28-source pre-training pool: removed duplicate archives, excluded evaluation datasets, filtered unsuitable resolutions, converted grayscale images to RGB, and created WebDataset shards.
- Implemented bounded multi-scale cropping and the DINO-style student–teacher objective with EMA updates, initialized from DINOv3 ViT-Base/16.
- Implemented LP-FT, LLRD ( $\gamma = 0.75$ ),  $256 \times 256$  inputs, and horizontal-flip TTA; organized the training configurations and scripts.
- Ran the main MedicinalPlant transfer experiments, tracked training curves, and prepared the pipeline and training-dynamics figures.

## Writing and analysis

I drafted the abstract, Introduction, Methodology, and portions of the Discussion. I wrote the corpus-audit and training descriptions, prepared figure captions, checked notation and reported values, and helped interpret the cumulative ablation without presenting it as independent causal evidence.

## Shared work

With Md Mehedi Hasan, I selected the research questions, reviewed the literature, checked the final numbers, edited the complete paper, and verified the final LaTeX/PDF output.